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A Co-evolutionary Genetic Programming Framework for Market-Adaptive Formulaic Alpha Generation

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Abstract: This paper develops a Co-evolutionary Genetic Programming (CEGP) framework to construct market-adaptive alpha factors in dynamic market environments. We employ a Hidden Markov Model (HMM) to identify market and industry regimes daily based on technical indicators. Subpopulations of factors evolve independently, each dedicated to a specific market regime identified by the HMM. We design a structure-aware feedback mechanism that quantifies the relative importance of each subpopulation and incorporates the feedback into the fitness evaluation of individual programs to align factor evolution with portfolio performance objectives. Finally, these specialized sub-factors are dynamically integrated into a market-adaptive composite factor through a gating mechanism considering the regime-specific weights derived from HMM. Empirical analysis on CSI 300 constituents demonstrates that our proposed factor generates significant alpha and exhibits strong robustness across diverse market regimes. The Top-k portfolios display a clear stratification effect under both long-term holding and short-term rotation strategies, validating the method's efficacy in return ranking, risk resilience, and regime adaptability. By integrating the interpretability and theoretical grounding of traditional factor engineering with the flexibility and automation of heuristic search, our framework contributes, both in theory and practice, to the ongoing shift in quantitative financial modeling from static towards dynamic, adaptive factors.

Keywords: Computational Finance; Factor Investing; Synergistic Formulaic Alpha Generation; Genetic Programming; Hidden Markov Model

1. Introduction

The stock market is a highly complex and dynamically evolving system, characterized by nonlinear interactions and heterogeneity across time, sectors, and participants. A vast body of financial literature, rooted in asset pricing theory, has established that the cross-section of stock returns can be systematically explained by exposure to a set of common risk factors. Seminal works, such as the Fama-French three-factor model (Fama and French, 1993) and its subsequent extensions (Carhart, 1997; Fama and French, 2015), demonstrate that factors like market, size, value, momentum, profitability, and investment capture a significant portion of return variations. This theoretical foundation has motivated both academics and practitioners to seek out new “alpha factors”—predictive signals that can generate excess returns beyond these common risk exposures. Traditional quantitative factor libraries, such as the Alpha101 (Kakushadze 2016), built upon economically or technically motivated heuristics, are prime examples of this pursuit and have become indispensable tools. However, the rapid evolution of financial markets has exposed critical limitations in traditional, statically designed factor construction. These challenges are particularly noteworthy and are well-documented in the empirical finance literature. First, static and manually designed rules struggle to capture time-varying regime dynamics, leading to the well-documented phenomena of performance instability and factor decay when market conditions shift (McLean and Pontiff, 2016). Second, handcrafted factors are inherently limited in scalability and in their ability to model the complex, nonlinear relationships that characterize financial markets, which are often state-dependent (Guidolin and Timmermann, 2007).

To address these deficiencies, recent research at the intersection of finance and computer science has explored heuristic approaches including Genetic Programming (GP) and other evolutionary algorithms. These methods can automatically generate factor expressions and flexibly model nonlinear dependencies, offering swift adaptability to market dynamics. Yet, purely algorithmically generated factors also present significant drawbacks: their structures often lack

economic interpretability, and they fail to explicitly incorporate regime information, which can result in fragility across market cycles.

Guided by the Adaptive Markets Hypothesis (Lo, 2004) which posits that market efficiency is dynamic and evolutionary, we propose a Co-evolutionary Genetic Programming (CEGP) framework to bridge this gap. It is explicitly designed to address the dual challenges of time-varying factor efficacy and market regime dependency by integrating regime identification with synergistic factor construction. In contrast to traditional factors that are treated as independent, static formulas whose value is primarily determined by their historical backtesting performance, the CEGP framework adopts a synergistic construction approach. Our method aims to build an ecological factor that is structured (composed of multiple sub-modules), regime-dependent (adapting its behavior to changing market states) and synergistic (where internal modules co-evolve by optimizing a collective objective). This hybrid approach retains the theoretical rigor and interpretability of traditional factor design while leveraging the scalability and adaptive search capabilities of heuristic methods. Specifically, this framework employs a Hidden Markov Model (HMM) constructed using volatility indicators and short-term return signals to identify multi-level regimes for both the market and industries. From these regimes, we derive probabilistic distributions that serve as soft labels and adaptive weights during factor construction. For each regime, a dedicated subpopulation evolves independently using genetic programming, thereby generating regime-specific subfactors. These subfactors are then integrated via a probabilistic gating mechanism, where regime probabilities dynamically govern their contributions to the composite factor.

The CEGP framework is designed to resolve the core tension between interpretability and adaptivity: sub-factors retain a clear economic or technical structure, while the gating mechanism ensures a dynamic response to changing environments. The framework is operationalized and validated through a three-step, multi-tiered experimental design. First, we conduct single-factor and decile-based portfolio simulations to assess the statistical characteristics, ranking efficacy, and robustness of both the composite factor and subfactors. Subsequently, we construct multi-factor portfolios to test whether the co-evolutionary mechanism enhances the effectiveness and complementarity of multi-regime factor integration. Finally, we perform regime-specific evaluations, measuring the contribution of each sub-factor to validate the dynamic design of our framework. A series of robustness checks are also performed to confirm the generalizability and practicability of the results.

The remainder of this paper is organized as follows: Section 2 reviews the related literature; Section 3 describes our data and regime modeling methodology; Section 4 introduces the CEGP framework; Sections 5 and 6 present the empirical results and robustness tests; and Section 7 concludes with implications and future research directions.

2. Literature Review

2.1 Factor Investing and Market Regimes

The quest to explain and predict cross-sectional stock returns is a cornerstone of financial economics, with factor investing emerging as a dominant paradigm. The foundational work of Fama and French (1992, 1993) established the eponymous three-factor model (market, size, value), demonstrating that these factors capture a significant portion of equity return variations. This framework was extended to include momentum (Jegadeesh and Titman, 1993; Carhart, 1997), and later, profitability and investment factors (Fama and French, 2015), forming the five-factor model. Beyond the U.S. market, these factors have been validated in global and international contexts (Griffin, 2002), underscoring their broad applicability in international finance. Recent research has further demonstrated the effectiveness of combined factor strategies in regional markets such as

Europe (Bermejo et al., 2021).

However, factor investing faces challenges of efficiency decay and regime dependency. McLean and Pontiff (2016) provided robust empirical evidence that the predictive power of discovered anomalies diminishes post-publication due to arbitrage activity. Furthermore, factor performance is profoundly regime-dependent, influenced by macroeconomic conditions and investor sentiment (Chordia et al., 2020; Gu et al., 2021). Studies have shown that well-designed multifactor portfolios must consider not only firm characteristics but also portfolio diversification effects to optimize performance (Heinrich et al., 2021). The existence of distinct market regimes—bull, bear, sideways—and their impact on asset pricing has been extensively documented (Guidolin and Timmermann, 2007). These regimes, often driven by global risk appetite and international capital flows, are a central concern in international asset pricing. Empirical analyses confirm the benefits of factor investing across different portfolio allocation strategies, though the optimal approach may vary depending on investment constraints and objectives (Bertrand and Prigent, 2025).

2.2 Computational Approaches to Adaptive Factor Generation

In response to challenges of factor efficiency decay and regime dependency, research at the intersection of finance and computer science has focused on automated and adaptive factor discovery. A foundational contribution in formulaic alpha generation was made by Kakushadze (2016). Subsequent advances leveraged evolutionary algorithms: Zhang et al. (2020) proposed AutoAlpha for hierarchical search and ensemble ranking, while Yu et al. (2023) and Shin et al. (2024) employed reinforcement learning to generate synergistic alpha collections.

Beyond single-factor generation, significant research has addressed the critical question of how to optimally combine multiple factors into portfolios. Blonk and Messow (2025) systematically compared integrated versus mixing approaches for constructing long-only multifactor credit portfolios, finding that integrated approaches generally yield superior risk-adjusted returns by avoiding exposure to bonds scoring poorly on individual factors. At the methodology level, Beaulieu et al. (2025) proposed identification-robust inference procedures for multifactor asset pricing models, addressing crucial statistical challenges in parameter estimation and hypothesis testing when factors may be elusive or model restrictions need testing. Furthermore, DeMIGUEL et al. (2024) advanced this domain by developing a conditional multifactor portfolio framework that incorporates volatility-managed strategies, demonstrating out-of-sample performance net of costs and showing that factor risk prices generally decrease with market volatility.

To enhance adaptability, the integration of regime-switching models like the Hidden Markov Model (HMM) has proven valuable. Palupi et al. (2021) used HMMs to classify market regimes, with predictive value demonstrated in diverse international markets (Dar et al., 2022; Soltani and Abbes, 2023). Gopinathan et al. (2024) advanced HMM modeling with Gaussian mixtures. Parallel to these model-based approaches, Genetic Programming (GP) has been used to simulate adaptive market behaviors (Chen et al., 2022; Li et al., 2024). Recent deep learning approaches have further enriched the field. Ma et al. (2025) developed Stockformer for the Chinese market, integrating wavelet decomposition and self-attention mechanisms. Liu et al. (2022) treated price charts as images using DLNNs, while Zhao et al. (2023) proposed a Time Series Relational Model (TSRM) combining LSTM with graph networks. Htun et al. (2023) systematically reviewed feature selection techniques in financial forecasting.

2.3 Research Gap and Contribution

Synthesizing these two streams of literature reveals a critical gap. While financial theory has firmly established the problems of factor decay and regime dependency, the computational literature has developed powerful tools (GP, HMM, RL) for automation and adaptation. However, many automated approaches lack a structured economic narrative, often generating "black-box" factors

that are difficult to interpret and may not explicitly internalize regime dynamics.

Our work seeks to bridge this gap. We integrate the theoretical underpinnings of regime-dependent asset pricing with the automated search power of co-evolutionary genetic programming. By constructing a structured, regime-aware factor model, we provide a transparent and adaptive solution to the problem of time-varying factor efficacy. Specifically, our contributions are threefold. First, we propose a novel framework that formally integrates regime identification (via HMM) with a co-evolutionary factor discovery process, ensuring factors are inherently adaptive to market states. Second, we introduce a structure-aware adaptive feedback mechanism, inspired by Shapley value theory, to quantitatively assess the contribution of each subfactor module. This mechanism dynamically adjusts their evolutionary pathways via a gating network, thereby steering the population toward more synergistic structures. Third, we demonstrate empirically that this framework generates factors that are both interpretable and performant, achieving a balance often missing in the literature.

Although our empirical application focuses on the Chinese market, a major component of global equity indices and of significant interest to international investors (Carpenter et al., 2021), the CEGP framework is designed as a generalizable solution that aims to mitigate the intertwined and pervasive challenges of factor decay and regime dependency in international quantitative finance.

3. Data Preparation and Market Regime Modeling

3.1 Data Sources and Fundamental Indicators

We select the constituent stocks of the CSI 300 Index, one of the most representative benchmarks in the Chinese stock market, as our empirical sample. The index comprises 300 large-cap and highly liquid stocks listed on the Shanghai and Shenzhen Stock Exchanges, effectively capturing the performance of the Chinese blue-chip universe and offering broad market coverage. The sample period spans from January 1, 2017, to February 15, 2025, encompassing multiple market phases including bullish, bearish, and sideways regimes. This enables a comprehensive evaluation of the proposed factor model across diverse market environments.

This study utilizes daily-level market data sourced from the Wind financial database. Each constituent stock is labeled with its industry sector based on the first-level classification of the Shenwan industry system, allowing subsequent modeling of industry-level regimes and structural heterogeneity. The daily returns of the CSI 300 Index serve as the market benchmark. Table 1 provides an overview of the key indicators, time series operations, and industry categorizations used in this study.

Table 1. Data, Formulas, and Industry Examples

Category	Examples
Operators	<i>Rolling-Mean, Rolling-Max, Minus, Multiply,...</i>
Industry	<i>Bank, Agriculture, Energy,...</i>
Features	<i>open, high, close, amount, pc, pb,...</i>
Constant	<i>-1,0,2,3,...</i>

Based on these fundamental fields, we further compute a range of derived financial indicators such as daily return (ret_{1d}), three-day rolling return (ret_{3d}), and five-day rolling return (ret_{5d}).

3.2 HMM-Based Market Regime Modeling

The inherent complexity and regime-dependent dynamics of financial markets often render static factors ineffective, making regime identification a foundational step for constructing adaptive

models. To address this challenge, our study employs a Hidden Markov Model (HMM) to establish a regime identification framework. The HMM captures latent market states from observable variables and models their probabilistic transitions. To account for the uncertainty in daily classification, we represent each day's market state not as a single label but as a smoothed posterior probability vector, providing a continuous and nuanced depiction of regime affiliation.

3.2.1 Observed Variables of HMM

Informed by both established financial literature and the objectives of our study, we select a comprehensive set of six observable variables to characterize the market dynamics. Our choice is designed to capture three complementary dimensions of price movements: trend, volatility, and strength. Specifically, the set comprises: (i) the 1-, 3-, and 5-day returns of the CSI 300 Index (ret_{1d} , ret_{3d} , ret_{5d}) to measure directional movement; and (ii) three technical indicators—the Relative Strength Index (RSI) for momentum, the Average True Range (ATR) for volatility, and the Average Directional Index (ADX) for trend strength. This multi-dimensional input $X_t = (ret_{1d,t}, ret_{3d,t}, ret_{5d,t}, RSI_t, ATR_t, ADX_t)$ is designed to jointly capture return momentum, sentiment dynamics, and structural intensity of the market and is crucial for the HMM to accurately delineate distinct market regimes. We present the details of these variables below.

(1) Return-Based Variables (ret_{1d} , ret_{3d} , ret_{5d})

Return is the most direct metric for assessing market trends. We compute the 1-day, 3-day, and 5-day returns of the CSI 300 Index to capture short-term directional movements and fluctuations in trend intensity over varying horizons:

$$ret_{nd,t} = \frac{P_t - P_{t-n}}{P_{t-n}}, \quad (n = 1, 3, 5).$$

P_t denotes the closing price of the CSI 300 Index on day t , and $ret_{nd,t}$ represents the n -day return of the index as of day t .

(2) RSI (Relative Strength Index)

The Relative Strength Index (RSI), introduced by J. Welles Wilder, is a momentum-based oscillator designed to identify overbought or oversold conditions in the market:

$$RSI_t = 100 - \frac{100}{1 + RSI_t} RSt = \frac{AvgGain_{t,n}}{AvgLoss_{t,n}}.$$

$AvgGain_{t,n}$ and $AvgLoss_{t,n}$ represent the average gains and losses over the past n trading days, respectively ($n = 14$ in this study). The RSI value ranges from 0 to 100, where a higher value indicates stronger buying pressure. Typically, an RSI below 30 is interpreted as an oversold condition, while a value above 70 suggests an overbought condition.

(3) ATR (Average True Range)

The Average True Range (ATR) is a technical indicator used to measure market volatility. It is defined as the moving average of the True Range (TR) over a specified time window:

$$TR_t = \text{Max}(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|), \quad ATR_t = \frac{1}{n} \sum_{i=0}^{n-1} TR_{t-i}.$$

H_t , L_t and C_t represent the high, low, and closing prices on day t , respectively, and n denotes the length of the rolling window. A higher ATR indicates greater market volatility.

(4) ADX (Average Directional Index)

The Average Directional Index (ADX) measures the strength of a price trend. Its value ranges from 0 to 100, where ADX values below 20 are typically interpreted as indicating a weak trend, and values above 40 are considered to reflect a strong trend. The calculation of ADX involves

constructing the Positive Directional Indicator (+DI) and the Negative Directional Indicator (-DI), followed by the computation of the Directional Movement Index (DX). Then, ADX is obtained by applying a smoothing process to the DX series:

$$ADX_t = EMA_n(DX_t), DX_t = \frac{|+DI_t - (-DI_t)|}{+DI_t + (-DI_t)} \times 100.$$

The above three technical indicators collectively compensate for the informational limitations of return-based variables in identifying market structures, thereby enhancing the comprehensiveness of regime modeling.

3.2.2 HMM Model Specification and Parameter Estimation

To effectively identify the regime evolution process in the stock market, this study adopts the Hidden Markov Model (HMM) as the core tool for regime classification and dynamic inference. HMM is a classical sequential probabilistic graphical model that captures the transition dynamics between unobservable hidden states, while simultaneously incorporating observable variables for step-by-step regime identification. Its capabilities in smooth probabilistic modeling and outputting state transition probabilities make it particularly well-suited for characterizing the complex dynamics of financial markets, where regime boundaries are often blurred and transitions uncertain.

For the model specification, we define the hidden state space of the HMM to consist of three latent regimes, denoted as $S_t \in \{0,1,2\}$. These latent states are not assigned predefined economic interpretations at the outset. Instead, after model estimation, the characteristics of each state are identified by analyzing the mean and distribution of the observable variables under each regime, along with their corresponding return, volatility, and trend properties, in order to infer the economic meaning of each market state.

The HMM consists of two core components: the state transition mechanism that describes the temporal evolution of hidden states through a state transition probability matrix, and the observation generation mechanism that models the joint distribution of observable variables conditional on each hidden state. Given that the observed sequence comprises continuous variables, we adopt a Gaussian mixture observation model. Specifically, for any hidden state S_t , the observation vector X_t is assumed to follow a multivariate Gaussian distribution characterized by a mean vector μ_s and a covariance matrix Σ_s . Under this formulation, each market regime is represented by a distinct statistical cluster, and the differences among regimes are reflected in the joint distributional structure of returns and technical indicators.

For each trading day t , the historical observation sequence up to day t , $\{X_1, X_2, \dots, X_t\}$ is used for model construction and regime probability inference. The objective of our HMM is not model fitting as in traditional supervised learning, but rather sequential inference of the hidden state's posterior distribution. Instead of optimizing classification accuracy over a training set, the model's goal is to maximize the likelihood of the observation sequence by leveraging its local statistical characteristics. Specifically, the HMM is trained using data prior to 2017. For each trading day after January 1, 2017, the posterior probability distribution over the hidden states is derived using Bayes' theorem, denoted as $P(S_t = s | X_{1:t})$. This results in a three-dimensional vector $W_t^{mkt} = [W_{mkt,t}^{(0)}, W_{mkt,t}^{(1)}, W_{mkt,t}^{(2)}]$, representing the probabilities of the market being in state 0, state 1, or state 2 on that day. This probability vector can be used not only to assign the most likely state by selecting the one with the highest probability but also as a set of continuous weights for use in the subsequent construction of structured factors.

After completing the Hidden Markov Model (HMM) estimation and regime identification, this study conducts a systematic analysis of the distributional characteristics of the observable variables under each inferred regime in order to assign economic meaning to the three latent states produced

by the model. Although the HMM only requires specification of the number of latent states during model initialization and does not impose any prior economic interpretation, the post-estimation mapping between regimes and observations allows for the backward identification of the corresponding market conditions associated with each state. We assign each observation point in the HMM framework a latent regime label, denoted as $\hat{S}_t \in \{0, 1, 2\}$, and use this classification to group the observation variables accordingly. Table 2 presents the results of comparative statistical tests for return-related and trend-related indicators under each regime. The results reveal substantial differences in market characteristics across the three latent states.

Table 2. Descriptive Statistics of Observed Variables Across Market Regimes

Variable	State 0	State 1	State 2	F-value	KS(MAX)
ret_{1d}	-0.002(0.028)	0.006(0.014)	-0.003(0.012)	24.86***	0.204***
ret_{3d}	-0.008(0.046)	0.020(0.021)	-0.008(0.017)	80.66***	0.393***
ret_{5d}	-0.012(0.059)	0.031(0.025)	-0.012(0.021)	129.60***	0.522***
RSI	-0.417(0.837)	0.941(0.631)	-0.557(0.723)	446.46***	0.601***
ATR	1.206(0.969)	-0.335(0.551)	-0.620(0.291)	450.16***	0.533***
ADX	0.481(1.006)	-0.274(0.947)	-0.115(0.908)	5.22**	0.124***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. F-values from one-way ANOVA; KS represents maximum pairwise Kolmogorov-Smirnov statistic.

Based on the statistical results reported in Table 2, State 1 is characterized by the highest average values for short-term return indicators (ret_{1d} , ret_{3d} , ret_{5d}), a significantly positive RSI (mean = 0.941), and relatively moderate levels of ATR (−0.335) and ADX (−0.274). These patterns indicate a market condition with moderate volatility and strong upward momentum, which can be interpreted as a short-term upward-trending regime. In contrast, State 0 is associated with negative return indicators, with ret_{3d} and ret_{5d} showing mean values of −0.008 and −0.012, respectively. The volatility and trend indicators, ATR (1.206) and ADX (0.481), are at elevated levels, suggesting that the market is experiencing sharp fluctuations and downward pressure. This state is best interpreted as a short-term downward-trending regime. State 2 also exhibits negative returns, but with the lowest average RSI (−0.557) and relatively low levels of ATR and ADX . This indicates a low-volatility, weak-trend market environment, consistent with a range-bound or neutral regime.

Regarding statistical significance, the ANOVA test shows that RSI ($F = 446.5$), ATR ($F = 450.2$), and ret_{5d} ($F = 129.6$) exhibit the most significant differences across the three regimes, identifying them as key variables for distinguishing market states. The Kolmogorov–Smirnov (KS) tests further confirm that the distributional differences of each variable across regime pairs are statistically significant, indicating that the HMM-derived state labels exhibit strong discriminatory power in terms of observed variable distributions.

We use $W_{mkt,t} = [W_{mkt,t}^{(0)}, W_{mkt,t}^{(1)}, W_{mkt,t}^{(2)}]$, the posterior probability distribution vector for the market regime on each trading day, as a proxy variable for the market condition on the following day. This design is consistent with the “T+1” trading mechanism in the Chinese stock market and aligns with the trading strategy setup in the subsequent sections of this study. Compared to hard classification, this soft-labeling approach provides greater smoothness and flexibility, reducing instability caused by abrupt regime switching and improving the reliability of strategy execution.

3.2.3 Industry Regime Modeling and Confidence-Based Weighting

To further capture the structural dynamics at the industry level, this study extends the market regime modeling and constructs an HMM for each first-level industry indices under the Shenwan classification system. Specifically, we aggregate each industry’s corresponding technical indicators, including RSI_{ind} , ATR_{ind} and ADX_{ind} , along with return indicators $ret_{ind,1d}$, $ret_{ind,3d}$ and $ret_{ind,5d}$, and

train a unified HMM for industry-level regime identification. Consistent with the market regime model, the number of hidden states for the industry model is also set to three. For each industry $d \in D$, the HMM outputs a daily posterior probability vector, denoted as $W_{ind,t} = [W_{ind,t}^{(0)}, W_{ind,t}^{(1)}, W_{ind,t}^{(2)}]$, which represents the probabilities that industry d on day t belongs to each of the three latent states. By aligning the industry classification with the stock-level industry labels, we map each stock's daily industry regime vector accordingly. This mapping constructs the industry-level regime input features for each individual stock.

Once the daily posterior probabilities of market and industry regimes are obtained, each stock on each trading day is assigned two state transition probability vectors, corresponding to the market-level and industry-level regimes. In financial markets, stock prices are influenced by both market-wide and industry-specific dynamics. However, the structural impact of market and industry regimes may vary across time periods and is often asymmetric or uncertain. Traditional regime classification methods typically rely on selecting the most likely state based on the maximum posterior probability. However, this approach does not account for the degree of concentration in the posterior distribution. For example, a posterior vector of $[0.9, 0.05, 0.05]$ indicates a much higher level of confidence than a more uniform distribution such as $[0.4, 0.3, 0.3]$.

To account for varying degrees of certainty in our daily regime classifications, we develop a dynamic, confidence-based weighting mechanism. The spread of the posterior probability vector (the difference between its maximum and minimum values) is used to indicate the confidence level, and a larger spread is translated into a higher weight. This logic is implemented by constructing two normalized confidence indicators, $p_m(t)$ and $p_i(t)$, which quantify the relative certainty of the market and industry regimes and directly determine the final weight allocation, as computed below:

$$\Delta mkt,t = \max(W_{mkt,t}) - \min(W_{mkt,t}), \Delta ind,t = \max(W_{ind,t}) - \min(W_{ind,t}),$$

$$p_m(t) = \frac{\Delta mkt,t}{\Delta mkt,t + \Delta ind,t + \varepsilon}, p_i(t) = 1 - p_m(t).$$

This formulation can be interpreted as a confidence-based weighting scheme. When the market regime is identified with higher certainty and the industry regime appears more ambiguous, greater weight should be assigned to the market regime, and vice versa. Unlike directly assigning hard-coded weights or using simple averaging, this structure offers a data-driven and relatively stable method for dynamic weight allocation. It distinguishes and quantifies the relative influence of market and industry regimes within the factor construction process.

4. Co-evolutionary Genetic Programming Framework

4.1 Genetic Programming Framework for Factor Discovery

Although deep learning models have demonstrated powerful predictive performance, their lack of interpretability significantly hinders trust establishment and effective risk management, both of which are essential in investment. To address this performance-interpretability trade-off in quantitative finance, we propose a novel framework for constructing structurally interpretable factors using co-evolutionary genetic programming (CEGP). This framework is designed to automatically discover high-performing alpha factors that adapt to changing market regimes.

In our CEGP framework, each factor is represented as a program expression tree composed of financial variables, such as open, high, and close prices, combined with functions from a predefined set. This set includes time-series operators (for example, RollingMean and RollingMax) and basic arithmetic operators. An initial population of expression trees is randomly generated. These candidate programs then undergo iterative improvement through genetic operations including mutation, crossover, and reproduction. The evolutionary process is guided by a fitness function that evaluates each expression based on its ability to predict future returns. Through this iterative

optimization, the algorithm gradually evolves more effective and interpretable factor structures. Table 3 summarizes the parameter settings used in the standalone evolution phase of the genetic programming algorithm.

Table 3. Parameter Settings for the Genetic Programming Algorithm

hall_of_fame	population_size	tournament_size
10	100	30
stopping_criteria	init_depth	p_crossover
1.0	(2,7)	0.3
p_subtree_mutation	p_hoist_mutation	generations
0.1	0.2	5
p_point_mutation	p_point_replace	n_components
0.2	0.5	10

4.2 Construction of Market-Adaptive Factor Structures

In financial markets, the performance of traditional alpha factors often relies heavily on firm-level trading data and technical indicators. However, in the Chinese stock market, broader market conditions such as the strength of the overall trend, the level of volatility, and the interaction between industries also play an important role in shaping asset price movements. Therefore, using conventional genetic programming methods without structural guidance may lead to the generation of factors that are structurally entangled and ineffective under varying market regimes. To this end, we introduce a structurally constrained expression format that embeds regime-specific logic directly into the factor design. This approach enables the generated factors to exhibit a clear dependence on the prevailing market environment. The final factor is constructed as a weighted combination of sub-factors, where each component corresponds to a particular market or industry regime:

$$H = p_m(W_{Market}^{(0)} \cdot A + W_{Market}^{(1)} \cdot B + W_{Market}^{(2)} \cdot C) + p_i(W_{Ind}^{(0)} \cdot D + W_{Ind}^{(1)} \cdot E + W_{Ind}^{(2)} \cdot F).$$

The structure consists of two main components that correspond to the overall market regime and the industry-specific regime, respectively. In this formulation, p_m and p_i represent the market regime weight and the industry regime weight. These weights are derived from the hidden state probabilities generated by the Hidden Markov Model (HMM), which is constructed using short-term return indicators and observable variables such as *RSI*. The weights reflect the extent to which an individual stock is influenced by market-wide and industry-specific trends at a given point in time. Variables $W_{market}^{(0)}$, $W_{market}^{(1)}$, $W_{market}^{(2)}$, $W_{ind}^{(0)}$, $W_{ind}^{(1)}$, and $W_{ind}^{(2)}$ denote regime-specific transition probabilities and serve as gating parameters that adjust the influence of each sub-factor under different market conditions. Sub-factors A , B , C , D , E , and F are generated by six independently evolving subpopulations. Each subpopulation focuses on discovering expression structures that perform better under a specific market or industry regime. To ensure consistency in magnitude, all sub-factors (A through F) are standardized using z-score normalization based on a 30-day rolling window. As a result, the final composite factor H is capable of dynamically adjusting its internal structure according to current market conditions. This allows the factor to be sensitive to trends and adaptive to regime changes. The overall model structure is illustrated in Figure 1.

Coefficient (RankIC), which is defined as follows:

$$Fitness_i = \frac{Mean[\rho(f_i(t) \times m_i(t), y(t))]}{Std[\rho(f_i(t) \times m_i(t), y(t))]},$$

where $f_i(t)$ denote the factor value on day t , $m_i(t)$ represent the regime transition probability label associated with the subpopulation to which the factor belongs, and $y(t)$ denote the realized return for that day. The function ρ corresponds to the RankIC function.

Then, in the structural combination stage, the best-performing individual from each subpopulation is combined according to the above structure to form the complete factor H , and its overall fitness is computed as $Fitness_H$. Finally, during the fitness feedback stage, we employ a contribution-aware metric based on the Shapley value to rigorously quantify the role each sub-factor plays in the overall performance of the composite factor H . Let the total fitness function be denoted by $F(H)$, and let the optimal combination of sub-factors be represented by $\{f_A, f_B, f_C, f_D, f_E, f_F\}$. The structural contribution of the i -th sub-factor is calculated approximately as:

$$\varphi_i \approx \sum_{S \subseteq C \setminus \{f_i\}} \frac{|S|!(n - |S| - 1)!}{n!} \cdot [F(S \cup \{f_i\}) - F(S)] (n = 6, i = A, B, C, D, E, F),$$

where S denotes any subset of sub-factors that does not include the i -th component, $F(S)$ represents the fitness of the composite factor constructed solely from the sub-factor subset S , and φ_i denotes the contribution of the optimal individual from the i -th subpopulation within the current structure. We employ a full enumeration strategy, computing all possible permutations of sub-factor subsets to accurately assess the marginal contribution of each subpopulation to overall performance. Subsequently, the structural contribution is fed back to the individual level within each subpopulation in order to adjust fitness evaluations. Specifically, the adjusted fitness of the j -th individual in a given subpopulation is defined as $\tilde{f}_{i,j} = f_{i,j}^{raw} + \alpha \cdot \varphi_i$, where $f_{i,j}^{raw}$ denotes the original fitness, and $\alpha \in [0, 1]$ represents the strength of the structural feedback, which is set to 0.5 in this study. This structural adjustment mechanism ensures that even if an individual does not exhibit strong local performance, it can still receive a high survival probability and evolutionary opportunity as long as its corresponding subpopulation contributes meaningfully to pattern recognition or trend detection within the composite factor structure.

In each round of factor generation, approximately 2,000 structurally diverse factor individuals are produced using the co-evolutionary genetic programming algorithm. To ensure the quality and stability of downstream stock selection strategies, we rank all generated factors based on their fitness scores on the training set, measured by fitness, which captures return-sorting effectiveness. The top 200 factors are then selected to form the final factor set, referred to as *Alpha CEGP*. The chosen scale is comparable to those of widely used quantitative factor sets such as *Alpha101*, *Alpha158*, and the Guotai Junan 191-factor library. This selection maintains diversity while mitigating the risk of overfitting. Furthermore, the structural co-evolution mechanism employed in this study ensures a comprehensive global exploration of the factor space. The top 200 factors have undergone multiple iterations and have accumulated sufficient expressive power and sensitivity to market trends, making them both representative and interpretable. Therefore, the number of selected factors does not exhibit a linear dependence on the total number of generated individuals. Rather, it reflects a compressed representation of high-performing structures and a refined consolidation of strategy signals. Our proposed framework is not a complete black-box model. Through its hierarchical design, it provides two levels of interpretability. At the macro-level, although the final composite factor structure is complex, its internal working mechanism is transparent; analysts can clearly identify which regime-specific sub-factors are active under particular market conditions. At the micro-level, since each sub-factor is a symbolic expression generated by genetic programming, it is inherently more transparent than the opaque weight matrices found in neural networks. This allows researchers to directly analyze these formulas to understand their underlying economic or technical

logic.

5. Simulation Experiments

5.1 Experimental Design

To evaluate the practical effectiveness of the proposed co-evolutionary structured factor framework, we design a series of benchmark and comparative experiments in this section. All experiments are conducted using daily data of the CSI 300 Index constituent stocks in the Chinese stock market, with the backtesting period spanning from January 2023 to February 2025. This time frame covers multiple market cycles and structural shifts.

In terms of strategy specification, we adopt two primary evaluation approaches: the Top- k stock selection strategy and the factor win rate (*WinRate*) metric, which jointly assess both return levels and stock-picking effectiveness.

In the Top- k strategy, we construct a stock portfolio at the end of each trading day based on the factor values, and open positions at the next day's market open, holding them for n consecutive trading days. This forms a rolling investment framework with a fixed holding horizon. Given that the effectiveness of our factors is intrinsically linked to their regime-weighted composition, we implement full turnover (i.e., a 100% replacement rate) in the Top- k strategy. This ensures that as regime weights update, the actual holdings are promptly adjusted, thereby enhancing responsiveness and avoiding unstructured accumulation of outdated factor signals.

For benchmarking purposes, we adopt an equal-weighted passive holding strategy as the reference. In each rebalancing period, all stocks within the sample (i.e., CSI 300 constituents) are treated as the investment universe and allocated equal capital weights. That is, we assume an equally weighted position across all eligible stocks on each rebalance date, serving as a baseline to assess the excess return and risk-adjusted improvement delivered by active stock selection strategies.

Additionally, since our factor structure already includes gating mechanisms and a standardization system to normalize factor magnitudes, no further scaling is applied in the multi-factor evaluation phase. The daily factor values across all stocks are aggregated with equal weights across sub-factors, and stock rankings within each rebalancing cycle are determined based on these aggregated scores.

Specifically, for each factor or factor ensemble, we sort all stocks on trading day T in descending order of their (weighted) factor values. The top k ranked stocks are selected to form the portfolio. The return on trading day T is then calculated as:

$$R_T^{top-K} = \frac{1}{k} \sum_{t=0}^n \sum_{j=1}^k r_{j,T+t+1},$$

where variable $r_{j,t+1}$ denotes the return of stock j , measured as the percentage change from the opening price on trading day $T+t+1$ to the opening price on trading day $T+t+2$. The strategy is updated on a rolling basis at each rebalancing interval, which simulates a short-term, factor-driven trading process.

To assess the stock selection effectiveness of single factor within the factor set, we calculate the *WinRate* metric as the proportion of factors that achieve positive cumulative returns relative to the CSI 300 Index over the backtesting period:

$$\text{WinRate} = \frac{N_{\text{number of factors achieving higher returns than CSI 300 Index}}}{N_{\text{total number of factors}}}.$$

We also conduct statistical analysis on Cumulative Return, Annualized Return, Annualized Volatility, Sharpe Ratio (with risk-free return set to zero in this experiment), and Maximum Drawdown. This set of metrics provides a consistent and fair basis for comparing the performance

of different factors and factor combinations. Sortino Ratio and the Calmar Ratio are calculated as additional risk-adjusted performance metrics:

$$SortinoRatio = \frac{R_p}{\sqrt{\frac{1}{N} \sum_{t=1}^N \min(0, R_t)^2}}, \quad CalmarRatio = \frac{R_p}{MaxDrawdown}.$$

SortinoRatio measures the excess return earned per unit of downside risk. It considers only negative volatility, making it more suitable for return distributions that exhibit skewness. *CalmarRatio* evaluates the amount of annualized return earned per unit of maximum drawdown. These indicators offer a more nuanced evaluation of downside risk and capital preservation.

5.2 CEGP Model Factor Distribution and Sub-Factor Decomposition

In the designed factor structure of this study, the final composite factor is composed of multiple sub-factor modules that are independently optimized and correspond to different market or industry regimes. Specifically, the structure covers two dimensions, namely the market level and the industry level, with three sub-expressions embedded under each dimension, represented by six sub-factors labeled *A*, *B*, *C*, *D*, *E* and *F*. To evaluate the practical effectiveness of the coordination mechanism embedded in this structure, we propose a reverse decomposition approach. Each complete factor is decomposed into six sub-factor sets according to their source modules. Each element in a sub-factor set is defined as the product of one of the six sub-expressions in the *Alpha CEGP* factor structure *H* and its corresponding market or industry regime transition probability:

$$W_{Market}^{(0)} \cdot A, W_{Market}^{(1)} \cdot B, W_{Market}^{(2)} \cdot C, W_{Ind}^{(0)} \cdot D, W_{Ind}^{(1)} \cdot E, W_{Ind}^{(2)} \cdot F,$$

corresponding to *Alpha A* to *Alpha F*, respectively.

The *H* factors generated by the co-evolutionary process contain multi-level regime dependencies. Without decomposition, it is difficult to interpret the contribution of each component under different market conditions. In subsequent experiments, decomposed backtesting allows us to evaluate the marginal impact of each sub-factor within its corresponding regime and to examine whether coordinated synergy among the components is achieved. Each sub-factor set has a clearly defined origin, structural formulation, and its relationship with regime weights. We conduct Top-*k* strategy backtests and *WinRate* analysis for both the primary *Alpha CEGP* factor set and the six decomposed sub-factor sets. This enables us to examine whether the preservation of structural integrity under the co-evolutionary and regime-weighted mechanism leads to significant improvements in overall predictive performance.

5.3 Single-Factor Performance Evaluation

In this experiment, we aim to capture the short-term effectiveness of factor signals as well as the dynamic rhythm of market regime changes. To this end, we adopt a Top-20 stock selection strategy with a daily rebalancing cycle. The evaluation focuses on the primary factor set, *Alpha CEGP*, generated by the co-evolutionary genetic programming model. We conduct a distributional comparison of various performance metrics for single-factor strategies. The baseline is defined as the return from an equal-weighted portfolio of all CSI 300 Index constituents. The results are reported in Table 5-1.

Table 4. Single-Factor Performance of *Alpha CEGP* ($k = 20, n = 1$)

<i>Alpha CEGP</i>	<i>AnnualReturn</i>	<i>SharpeRatio</i>	<i>MaxDrawdown</i>	<i>SortinoRatio</i>	<i>WinRate</i>
mean	0.1803	0.7219	-0.2263	1.3161	69%
std	0.1216	0.3669	0.0456	0.6786	
min	-0.1359	-0.4018	-0.3890	-0.6406	

25%	0.1036	0.4956	-0.2522	0.8898	
50%	0.1716	0.7273	-0.2181	1.3465	
75%	0.2486	0.9504	-0.1945	1.7412	
max	0.5376	1.6246	-0.1426	2.8971	
count	200	200	200	200	138
baseline	0.1176	0.6001	-0.1828	1.0977	

In the evaluation of single-factor performance, we conduct a systematic analysis of 200 factors generated by the co-evolutionary genetic programming model and compare their overall statistical characteristics with those of traditional benchmark factors. The results indicate that *Alpha CEGP* significantly outperforms the benchmark across several key metrics. Specifically, the average annualized return of *Alpha CEGP* factors reaches 18.03 percent, which is substantially higher than the 11.76 percent observed for the benchmark. The average *SharpeRatio* of *Alpha CEGP* is 0.72, representing a notable improvement over the baseline value of 0.60, indicating stronger excess returns per unit of risk. Moreover, in terms of downside risk, the average *SortinoRatio* of *Alpha CEGP* is 1.32, exceeding the benchmark value of 1.10, further demonstrating the superior risk-adjusted returns of the proposed factor set. Although *Alpha CEGP* exhibits a slightly higher average maximum drawdown at -22.63 percent compared to the benchmark's -18.28 percent, the distributional characteristics suggest that most of the factors still maintain good control over tail risk. In addition, the win rate, defined under a predefined return threshold, reaches 69 percent, indicating that the CEGP framework exhibits strong robustness and broad applicability across different market environments.

5.4 Multi-Factor Strategy Performance

To comprehensively evaluate the performance of the proposed *Alpha CEGP* factors under different holding periods and stock selection sizes, we conduct a series of Top- k backtesting experiments. Specifically, we adopt the parameter setting ($k = 20$, $n = 1$) to test the robustness of the factor's stock selection ability under various operational scenarios. At each rebalancing interval, *Alpha CEGP* and the six decomposed sub-factors (*Alpha A* to *Alpha F*) are ranked based on their factor scores, and the top-ranked stocks are selected to construct the portfolio. These strategies are then compared to the benchmark strategy, which is defined as an equal-weighted portfolio of all CSI 300 constituents. The backtesting return curves are presented in Figure 2.

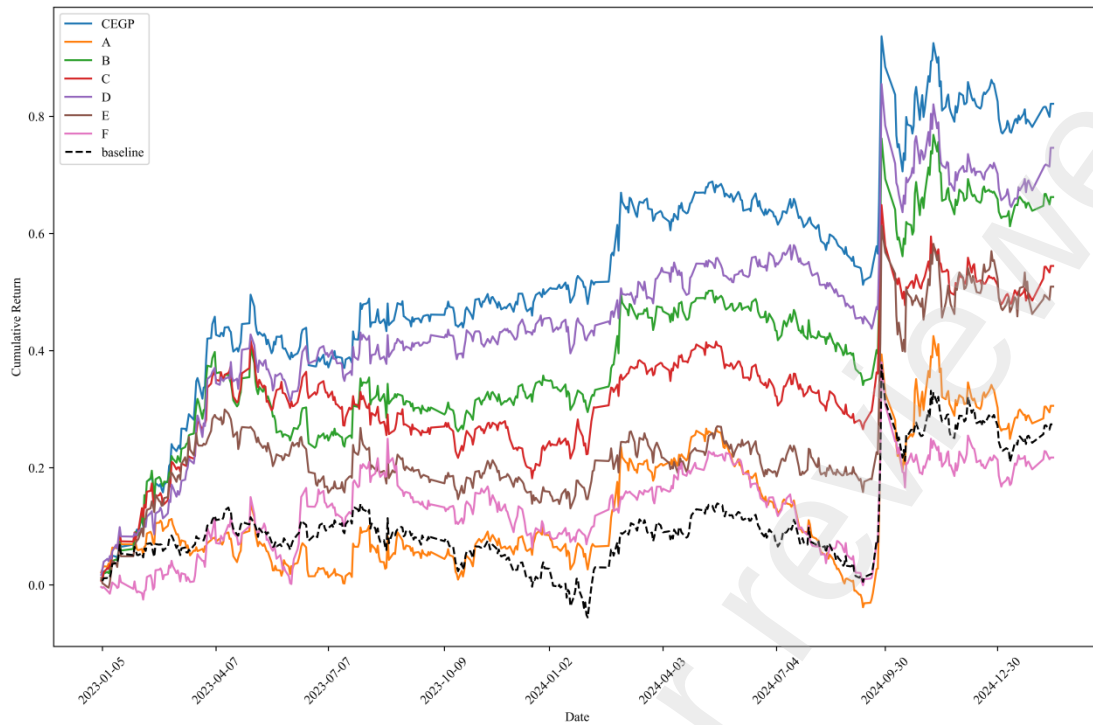


Figure 2. Backtesting Results of *Alpha CEGP* and Decomposed Sub-Factors ($k = 20, n = 1$)

From the cumulative return curves, *Alpha CEGP* exhibits a consistently upward trend under all three parameter configurations. It not only achieves significantly higher overall returns compared to the baseline but also maintains strong stability across different market environments and holding periods. The return curve of *Alpha CEGP* effectively forms the upper envelope of its sub-factors, indicating that the composite factor adaptively integrates the strengths of individual sub-factors across different periods. This reflects its superior adaptability and resilience to noise. Table 5 presents the backtesting statistics under the ($k = 20, n = 1$) configuration.

Table 5. Backtesting Results for *Alpha CEGP* and Sub-factors ($k = 20, n = 1$)

	<i>AnnualReturn</i>	<i>SharpeRatio</i>	<i>MaxDrawdown</i>	<i>CalmarRatio</i>
<i>Alpha CEGP</i>	0.4083(0.0188)	1.3690	0.2307	1.7699
<i>Alpha A</i>	0.1517(0.0176)	0.5443	0.3062	0.4954
<i>Alpha B</i>	0.3285(0.0181)	1.1428	0.2011	1.6339
<i>Alpha C</i>	0.2702(0.0157)	1.0832	0.2349	1.1499
<i>Alpha D</i>	0.3702(0.0181)	1.2918	0.2188	1.6922
<i>Alpha E</i>	0.2526(0.0188)	0.8462	0.2152	1.1742
<i>Alpha F</i>	0.1079(0.0161)	0.4226	0.2502	0.4212
<i>baseline</i>	0.1363(0.0143)	0.5989	0.1939	0.7027

Based on the overall results presented in Table 5, the *Alpha CEGP* factors demonstrates significantly superior aggregate return performance compared to all individual sub-factor sets. Its annualized return reaches 0.4083, the highest among all strategies, substantially outperforming the next best sub-factors, *Alpha D* (0.3702) and *Alpha B* (0.3285). The Sharpe ratio of *Alpha CEGP* is 1.3690, which also exceeds that of *Alpha D* (1.2918) and *Alpha C* (1.0832), indicating a more favorable risk-adjusted return profile. Furthermore, its *CalmarRatio* reaches 1.7699, the highest among all strategies, clearly outperforming the baseline and all sub-factor sets. This reflects the factor's ability to achieve higher returns while maintaining effective drawdown control.

From the detailed performance metrics, the *Alpha CEGP* factors constructed under the co-evolutionary structure achieve a more favorable balance across multiple dimensions, including return level, volatility control, and risk-return efficiency. Although some individual sub-factors exhibit strong performance on specific metrics, *Alpha CEGP* demonstrates more stable and consistent performance overall, suggesting stronger robustness and reliability. These findings provide evidence indicating that integrating multiple regime-driven sub-factors through a gating mechanism can enhance the stability and overall performance of the strategy, offering greater practical potential.

5.5 Performance in Grouped Portfolios

To further assess the cross-sectional discriminative power of the *Alpha CEGP* factors in predicting stock returns, we conduct a decile-based cumulative return analysis. Specifically, on each trading day, all stocks in the market are sorted based on their regime-weighted factor values and then divided into ten equal-sized groups (deciles). Decile 1 represents the bottom 10 percent of stocks with the lowest factor scores, while Decile 10 represents the top 10 percent with the highest factor scores. Within each decile group, we calculate the equal-weighted average return of the included stocks and accumulate these returns over the entire sample period. The resulting cumulative return curves allow for a visual comparison of the long-term performance differences among the decile groups. This method provides intuitive evidence of the factor's ability to explain the cross-sectional distribution of returns and reveals whether stocks in higher deciles consistently generate excess returns over the sample period. The return curves from this decile-based backtesting experiment are shown in Figure 3.

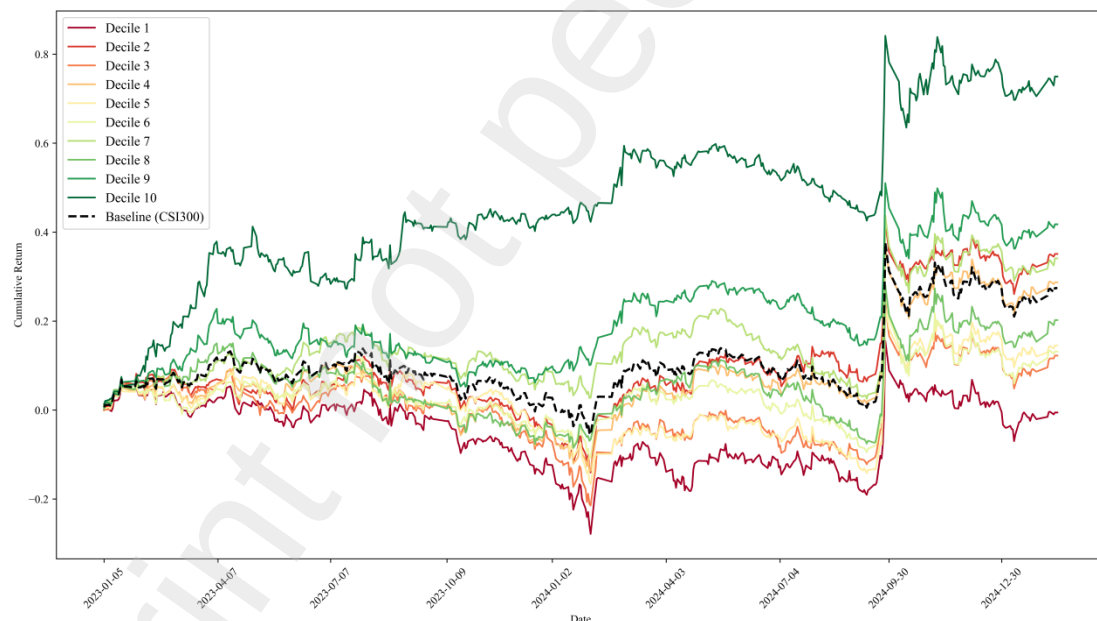


Figure 3. Decile-Based Backtesting Results of *Alpha CEGP* ($k = 20$, $n = 1$)

The experimental results show that the cumulative return curves across different decile groups exhibit a clear hierarchical structure. Although the performance differences among the middle to lower deciles are not highly pronounced, and some groups display similar return trajectories without the ideal strictly monotonic pattern, the Decile 10 group, which includes stocks with the highest factor values, consistently maintains a leading position throughout the sample period. Its cumulative return is significantly higher than that of all other groups. Decile 9 also demonstrates excess return characteristics, while Decile 1 consistently underperforms. These observations suggest that the factor values are effective in sorting stocks by return potential.

In conjunction with the single-factor experiments, the findings indicate that the *Alpha CEGP* factors stably generate excess returns under short-term holding strategies. To further investigate the possible reasons behind the observed group-wise performance, we introduce a marginal contribution evaluation mechanism based on portfolio performance changes. This mechanism quantitatively assesses the structural value of each subpopulation and feeds this information back into the individual-level fitness scores. Since the overall performance evaluation focuses on ranking quality, a sub-factor that consistently identifies high-return samples, even with limited global predictive power, can still exhibit a high marginal contribution within the structure. This mechanism leverages the nonlinear amplification effect of high-return stocks on overall performance, making sub-factors with strong sorting capability more likely to be retained and passed on to the next generation. Ultimately, this enhances the composite factor's ability to distinguish stocks within the Top- k range. These results suggest that the proposed factor structure is particularly well-suited for Top- k -oriented selection strategies and holds substantial potential for generating excess returns.

5.6 Marginal Contributions of Sub-Factor Sets under Specific Regimes

To further examine the unique contributions of each subpopulation within the co-evolutionary structure under its corresponding market regime, we design a marginal contribution experiment under specific regime conditions. Specifically, for each sub-factor component A to F within the *Alpha CEGP* composite factor, we remove it and construct a complementary factor set, denoted as *Alpha A'*, *Alpha B'*, *Alpha C'*, *Alpha D'*, *Alpha E'*, and *Alpha F'*. Each modified factor set is then evaluated under the regime dimension to which the excluded sub-factor originally belonged.

In terms of strategy implementation, we adopt a fixed stock selection rule with parameters ($k = 20, n = 1$). Each portfolio generates trading signals only under its corresponding regime, and the cumulative return under each specific regime is calculated accordingly. This approach enables us to identify the marginal impact of excluding individual sub-factors on the overall performance of the composite factor under the relevant market or industry condition. The results, in terms of strategy returns, are presented in Table 6.

Table 6. Backtesting Results for *Alpha CEGP* and Sub-Factor ($k = 20, n = 1$)

<i>Factor</i>	<i>Market 0</i>	<i>Market 1</i>	<i>Market 2</i>	<i>Industry 0</i>	<i>Industry 1</i>	<i>Industry 2</i>
<i>Alpha CEGP</i>	0.0128	0.0171	0.0017	0.1839	-0.1329	0.0424
<i>Alpha A'</i>	0.0052	0.0167	0.0025	0.1851	-0.1127	0.0385
<i>Alpha B'</i>	0.0134	0.0144	0.0029	0.1634	-0.1015	0.0361
<i>Alpha C'</i>	0.0127	0.0163	-0.0020	0.1508	-0.1299	0.0322
<i>Alpha D'</i>	0.0093	0.0150	0.0003	0.1401	-0.1025	0.0374
<i>Alpha E'</i>	0.0122	0.0201	0.0020	0.1687	-0.1399	0.0405
<i>Alpha F'</i>	0.0134	0.0181	0.0024	0.1821	-0.1045	0.0372

The results presented in Table 6 reveal that the complement portfolios exhibit significantly weaker performance under their corresponding regimes, as indicated by the bolded diagonal entries. For instance, *Alpha C'* yields a return of -0.0020 under market regime 2, which is substantially lower than the return of 0.0017 achieved by the complete *Alpha CEGP* factors, and is also the lowest value in that column. Similarly, other complement portfolios, such as *Alpha A'* and *Alpha B'*, also tend to produce the worst or second-worst returns under their respective regime conditions. This consistent pattern of performance deterioration indicates that each original sub-factor plays a critical role within its corresponding regime, contributing the discriminative power of the composite factor.

6. Robustness Test

6.1 Performance Comparison Under Different Holding Periods and Portfolio Sizes

To further evaluate the robustness of the proposed *Alpha CEGP* factor under varying parameter settings, we design two additional sets of strategy experiments. These experiments apply alternative configurations of the Top- k strategy, specifically $(k = 20, n = 3)$ and $(k = 30, n = 1)$. Figure 4 presents the results for the first setting, in which the holding period is extended to three trading days. This allows us to examine whether the factor's stock selection remains effective under a lower-frequency rebalancing schedule. Figure 5 shows the results for the second setting with 30 selected stocks to assess the performance differences when the selection scope becomes broader.

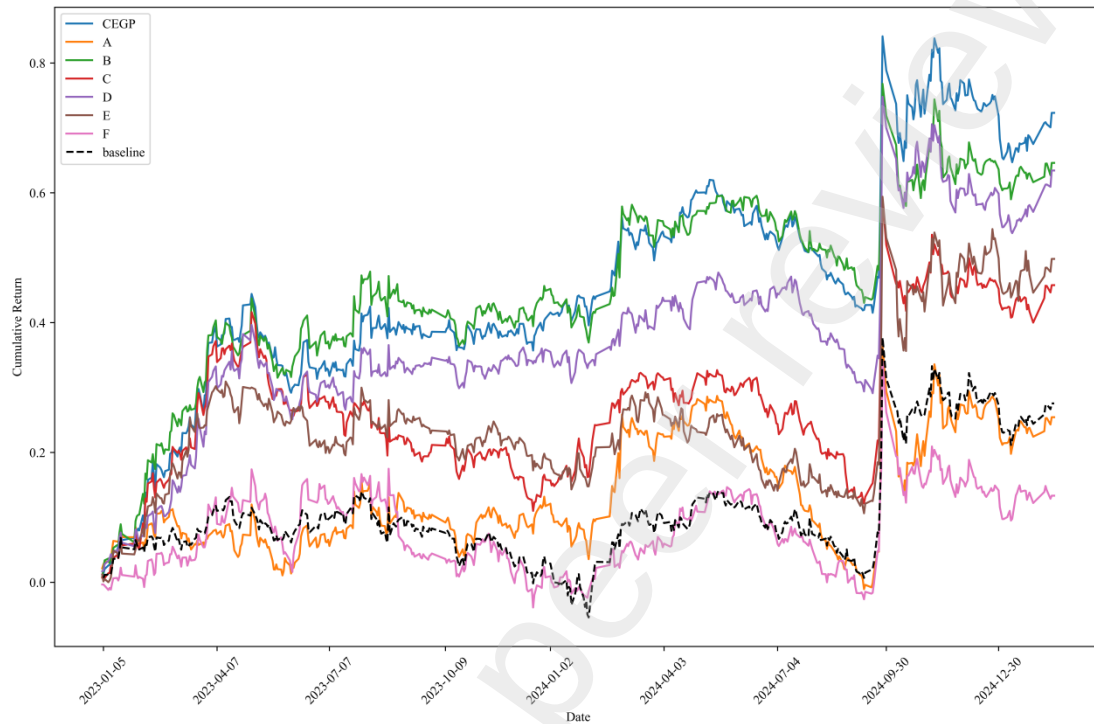


Figure 4. Backtesting Results for *Alpha CEGP* and Sub-Factors ($k = 20, n = 3$)

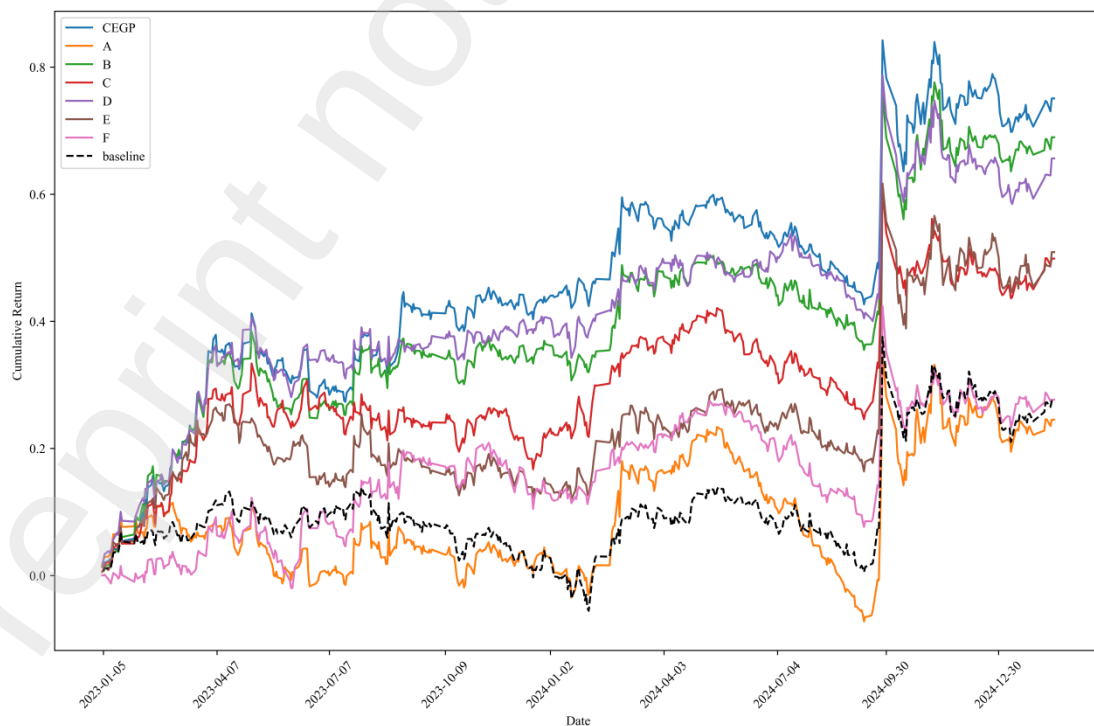


Figure 5. Backtesting Results for *Alpha CEGP* and Sub-Factors ($k = 30, n = 1$)

Figure 4 and 5 show that the *Alpha CEGP* portfolio consistently outperforms both the individual sub-factors and the baseline across both parameter settings. The comparative performance is stable, with the cumulative return curves exhibiting a smooth upward trajectory and well-controlled volatility. These findings indicate that the co-evolutionary factor constructed in this study remains effective not only under the baseline configuration but also under varying stock selection sizes and rebalancing frequencies, demonstrating strong adaptability across different strategy conditions.

6.2 Performance Comparison Using Randomized Stock Pools

To reduce the risk of overfitting associated with stock selection within the CSI 300 universe, we conduct a set of 10 randomized experiments. In each experiment, a subset of 200 stocks is randomly selected from the CSI 300 constituent pool and used as the investment universe. The same Top- k strategy as in the main analysis is applied, with parameters set to ($k = 20, n = 1$). The results are shown in Table 7.

Table 7. Summary Statistics of *Alpha CEGP* and Sub-Factors over 10 Randomized Stock Pools

Factor	Mean AnnualReturn	Mean SharpeRatio	Mean MaxDrawdown
<i>Alpha CEGP</i>	0.4242	1.4706	-0.1891
<i>Alpha A</i>	0.1184	0.4426	-0.2625
<i>Alpha B</i>	0.3662	1.3249	-0.1843
<i>Alpha C</i>	0.2538	1.0326	-0.1905
<i>Alpha D</i>	0.3600	1.3592	-0.2076
<i>Alpha E</i>	0.2696	0.9343	-0.2034
<i>Alpha F</i>	0.1528	0.6071	-0.1989
<i>baseline</i>	0.1372	0.6072	-0.1804

The results presented in Table 7 indicate that the *Alpha CEGP* factors demonstrate strong overall performance across multiple rounds of randomized stock pool tests. Its mean annual return reaches 42.42 percent, and the mean *SharpeRatio* is 1.4706, both significantly higher than those of the decomposed sub-factor sets and the baseline. Moreover, the maximum drawdown is limited to -18.91 percent, reflecting solid risk-adjusted returns and effective drawdown control. In contrast, the performance of individual sub-factors generally declines in the absence of the co-evolutionary structure. This is particularly evident in *Alpha A* and *Alpha F*, both of which show inferior return and risk metrics compared to *Alpha CEGP*. Even the better-performing sub-factors, such as *Alpha B* and *Alpha D*, fail to surpass *Alpha CEGP* in terms of either return level or stability. These findings suggest that the structured factors generated by the co-evolutionary framework remain robust when subjected to stock universe perturbations and sample variability.

7. Conclusion

This study introduces a Co-evolutionary Genetic Programming (CEGP) framework that successfully integrates market regime modeling with structured, heuristic factor discovery. By employing a Hidden Markov Model to identify market states and guiding the evolution of specialized sub-populations, our framework generates structurally interpretable alpha factors with significant, robust performance across diverse market conditions. Our empirical results confirm the superiority of this approach. The generated factors stably exhibit favorable stock selection ability under the Top- k strategy. The composite *Alpha CEGP* factor outperforms benchmarks and demonstrates clear synergistic gains over its individual components. Crucially, the structure-aware gating mechanism is validated through ablation studies, highlighting the positive marginal contribution of each regime-specific sub-factor and confirming the framework's adaptive

capabilities. Robustness tests under varying holding periods and alternative stock pool construction criteria further confirm the applicability and consistency of the *Alpha CEGP* factor across diverse market environments.

While our findings are promising, this study has several limitations that open avenues for future research. First, our regime identification relies on a three-state HMM with Gaussian emissions. Future work could explore alternative clustering techniques or more complex state-space models to capture different facets of market dynamics. Second, the universe of genetic programming operators and terminals, while comprehensive, was limited to traditional price and volume data. Integrating alternative datasets, such as textual news sentiment or macroeconomic variables, could further enrich the factor discovery process and potentially uncover novel sources of alpha. Finally, this study's empirical analysis was confined to the Chinese A-share market. Testing the framework's generalizability across international markets with varying microstructures and regulatory environments would be a valuable extension.

In conclusion, the proposed CEGP framework offers a balanced and powerful solution to the performance-interpretability trade-off in quantitative finance. By bridging the gap between the theoretical grounding of traditional factor investing and the automated search capabilities of evolutionary algorithms, our research provides both academic insights into regime-aware factor generation and an actionable tool for practitioners. This work contributes, both in theory and practice, to the ongoing and critical shift in quantitative modeling from static, monolithic factors towards dynamic, adaptive, and structurally transparent solutions.

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